INCH-POUND
MIL-DTL-24643/42D
1 October 2009
SUPERSEDING
MIL-DTL-24643/42C
22 August 2002

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -20 °C TO +90 °C, TYPE LS1SU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction, Non-Watertight

First	-	Copper conductor, tin-coated, ASTM B286, 20 AWG – 7 strands and 18 AWG, Class B stranding, ASTM B8 (7 by 0.0152 inches). For LS1SU-36, 32 AWG, Size 20 conductors and 4 AWG, Size 18 conductors. For LS1SU-60, 60 AWG, Size 20 conductors.
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation. Nominal wall thickness of 0.017 inch. Wall thickness may vary from nominal as necessary in order for completed cable to meet the specified electrical requirements.
Fourth	-	Braided shield of 34 or 36 AWG, coated or uncoated copper. Braid angle of 30 to 35 degrees; minimum coverage of 85 percent.
Fifth	-	Shield insulation of two polyester tapes, outer tape sealed. An alternate shield insulation may consist of one polyester tape, plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by Method 2. For LS1SU-36, the four 18 AWG conductors shall be numbered 1 through 4.
Sixth	-	The specified number of singles (see table I) shall be cabled together in accordance with 3.4.4 of MIL-DTL-24643. Cabling sequence shall be consecutive starting with no. 1 from the center outward. Fillers may be used as necessary to form a firm well-rounded assembly.
Seventh	-	An optional binder.
Eighth	-	Cross-linked polyolefin jacket (see table I for thickness).

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TABLE I. Details.

Military part no. M24643/42	Type and size	Number of conductors	Conductor size AWG no.	Cable jacket thickness min. avg. (inch)	Overall diameter	
					min. (inch)	max. (inch)
-01UN	LS1SU-36	4 32	18-7 20-7	0.075	0.910	0.985
-02UN	LS1SU-60	60	20-7	0.100	1.210	1.310

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic Electricals:

Conductor resistance (ohms/1000 feet at 25 °C, max.)	
ASTM B286, Size no. 20 – 7	10.93
ASTM B8, Size no. 18 (Class B)	7.05
Voltage withstand (volts, root mean square, min.)	
Conductor to component shield	2200
Component shield to component shield	500
Insulation resistance (megohms/1000 feet, min.)	
Conductor to component shield	500
Component shield to component shield	100
Conductor continuity	No failure
Shield continuity	No failure

Group A:

Visual and dimensional	No failure
Capacitance, 1 kilohertz (KHz) (pF/ft, max.)	72

Group B:

Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 200 °C)	50
Cold bending cable	No failure
Physicals (unaged)	
Insulation (extruded)	

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Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Group C:	
Physicals (aged) air oven	
Insulation (extruded)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Shrinkage	No failure
Shield (conformance to material, construction and coverage)	No failure
Group D:	
Flame propagation (cable)	No failure
Qualification Inspection:	
Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:	
Aging and compatibility (cable) (125±5 °C)	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Shield insulation	2
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2

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Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	.5
Fillers	5
Insulation	1.5
Durometer (jacket) - Type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LS1SU-36 and -60	1000

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2008-046)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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